

**Innovative Science Olympiad
Model Paper 2025 – 2026****Total Questions: 50****Time: 1.5 hr.****Total Marks: 100****Syllabus**

Matter in Our Surroundings, Is Matter Around Us Pure? Atoms and Molecules, Structure of the Atom, The Fundamental Unit of Life, Tissues, Improvement in Food Resources, Motion, Force and Laws of Motion, Gravitation, Work and Energy.

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. In a "Composite Fish Culture" system, five or six species of fish are grown in a single pond. What is the logic behind choosing specific species like Catlas (surface), Rohus (middle), and Mrigals (bottom)?
(A) So, they can fight and only the strongest fish survive for market.
(B) To ensure they all eat the same food so the farmer only buys one type of feed.
(C) To ensure that they utilize all the food available in all different zones of the pond without competing with each other.
(D) To ensure that the fish can cross-breed and produce a new hybrid species automatically.
2. A train starts from rest at a station and travels with a uniform acceleration of 2 m/s^2 . A passenger notices that the train passes a specific signal post exactly 10 seconds after starting. How much further will the train travel during the 11th second of its journey?
(A) 20 m (B) 21 m
(C) 121 m (D) 22 m
3. A student takes 10 mL of ethanol and dissolves it in enough water to make exactly 100 g of solution. What is the concentration of this solution expressed in volume by weight percent?
(A) 30 % (B) 10 %
(C) 1% (D) 20%
4. A student was observing two different slides under a microscope. Slide A showed cells with a large central vacuole and a distinct cell wall. Slide B showed cells with multiple small vacuoles and no cell wall. If both slides are placed in a highly concentrated salt

solution (hypertonic), what will be the most likely observation?

(A) Slide A cells will burst, while Slide B cells will shrink.

(B) Slide A cells will undergo plasmolysis, where the protoplast shrinks away from the cell wall, while Slide B cells will simply shrivel up.

(C) Slide B cells will become turgid, while Slide A cells will remain unaffected due to the cell wall.

(D) Both cells will expand because the salt solution provides nutrients for growth.

5. A car is traveling along a straight highway at a velocity of 30 m/s. The driver suddenly sees an obstacle and applies the brakes, causing a uniform retardation (negative acceleration) of 5 m/s². What is the total distance the car travels before coming to a complete stop?

(A) 45 m (B) 60 m
(C) 90 m (D) 180 m

6. Which of the following observations during Rutherford's gold foil experiment specifically led to the conclusion that the nucleus is extremely small compared to the total volume of the atom?

(A) Most alpha particles have not passed straight through the foil.

(B) Some alpha particles were deflected at small angles.

(C) Only about 1 in 20,000 alpha particles were thrown back.

(D) Alpha particles are positively charged

7. During a lab experiment, a section of a plant stem was stained and observed. The tissue showed cells with unevenly thickened walls at the corners, made of cellulose and pectin, with very little intercellular space. What is the primary function of this tissue?

(A) Providing buoyancy to aquatic plants to help them float.

(B) Providing mechanical strength and flexibility, allowing the stem to bend without breaking.

(C) Transporting water and minerals from the roots to the leaves.

(D) Conducting food prepared in the leaves to other parts of the plant.

8. Two athletes, Rohan and Sohan, are competing in a dash. On a displacement-time graph, Rohan's motion is represented by a line inclined at 30 ° to the time axis, while Sohan's line is inclined at 45 °. What can be concluded about their velocities?

(A) Sohan is accelerating while Rohan is moving at a uniform velocity.

(B) Rohan is running faster than Sohan.

(C) Both are running at the same speed but in different directions.

(D) Sohan is running faster than Rohan.

9. In the compound phosphorus pentachloride (PCl₅), what is the valency of the phosphorus atom?

(A) 3 (B) 4

(C) 5 (D) 2

10. Assertion: Xylem and Phloem are considered complex permanent tissues.

Reason: They are made up of more than one type of cell that work together as a unit to perform a common function.

(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true, but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true, but Reason is false.

(D) Assertion is false, but Reason is true.

11. An elevator moves up from the ground floor to the 10th floor, stops for 5 seconds, and then moves down to the 5th floor. If we plot a velocity-time graph for this entire journey, how is the displacement calculated?

(A) By adding the area above the time axis and subtracting the area below the time axis.

(B) By calculating the slope of the velocity-time graph at the 5-second mark.

(C) By taking the total area under the graph, treating all areas as positive.

(D) By finding the final velocity and multiplying it by the total time.

12. Why do wet clothes dry faster on a windy day even if the temperature remains the same?

(A) Wind increases the humidity of the surrounding air.

(B) Wind speed increases the kinetic energy of the water molecules.

(C) Moving air carries away water vapor, decreasing local humidity and increasing the rate of evaporation.

(D) Wind decreases the surface area of the clothes.

13. Assertion: Intercropping increases the total yield of the farm per unit area.

Reason: In intercropping, two or more crops are grown simultaneously in the same field in a definite pattern.

(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true, but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true, but Reason is false.

(D) Assertion is false, but Reason is true.

14. A courier van travels the first one-third of its route at 20 km/h the second one-third at 30 km/h, and the final one-third at 60 km/h. The average speed of the van is:

(A) 30 km/h (B) 36.6 km/h

(C) 40 km/h (D) 33.3 km/h

15. Identify the element that is named after Russia.

(A) Germanium (B) Ruthenium

(C) Polonium (D) Uranium

16. During a laboratory activity, students compared muscle tissue and nerve tissue. One student claimed both tissues conduct messages, while another said only one does. Which statement correctly explains their functions?

- (A) Both tissues conduct electrical impulses equally
- (B) Muscle tissue transmits messages, nerve tissue contracts
- (C) Nerve tissue transmits impulses, muscle tissue responds by contraction
- (D) Both tissues are involved only in movement

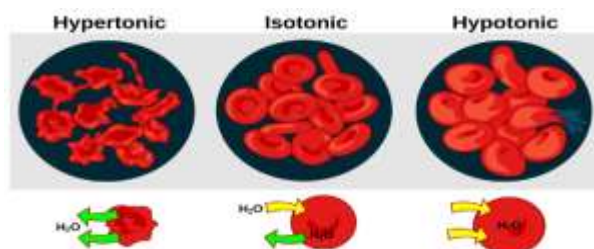
17. which statement about gravitational force is FALSE?

- (A) It only exists for the Earth and no other bodies.
- (B) It acts even without contact between bodies.
- (C) It is responsible for bringing objects thrown upward back to the ground.
- (D) It is always an attractive force.

18. If stability were attained with an octet (8 electrons), and an atom has the electronic configuration $1s^2 2s^2 2p^6 3s^2$ what is its most likely valency when it forms an ionic bond?

- (A) 1
- (B) 2
- (C) 6
- (D) 9

19.



The diagram shows red blood cells placed in three different solutions. In which condition will the red blood cell burst due to excess water entry?

- (A) Isotonic solution
- (B) Hypertonic solution

- (C) Hypotonic solution
- (D) Concentrated saline solution

20. Three students—Rahul, Sam, and Priya—carry identical 5 kg bags from the ground floor to the third floor (height 12 m). Rahul takes 20 seconds, Sam takes 30 seconds, and Priya takes 25 seconds. Who performed the most work?

- (A) Rahul, because he was the fastest.
- (B) Sam, because he spent the most time.
- (C) Priya, because her speed was average.
- (D) They all did the same amount of work.

21. A mixture of ammonium chloride and common salt is heated in a China dish covered with an inverted funnel. Which process allows for their separation?

- (A) Filtration
- (B) Distillation
- (C) Sublimation
- (D) Centrifugation

22. In an agricultural experiment, "Field A" was treated with chemical fertilizers (NPK) for five consecutive years, while "Field B" was treated with organic manure. What is the most likely long-term observation?

- (A) Field A will show a continuous increase in soil fertility and microbial activity.
- (B) Field B will eventually become barren due to a lack of concentrated nutrients.
- (C) Field A may show a decline in soil structure and a decrease in soil microorganisms over time.
- (D) There will be no difference in the soil quality of both fields.

- 23.** If the speed of a vehicle is tripled, its kinetic energy becomes:
 (A) 3 times (B) 6 times
 (C) 9 times (D) 27 times
- 24.** An unknown gas is observed to have no definite shape and no definite volume, but it can be compressed into a small volume in a cylinder. This property is exploited in ____.
 (A) Water cooling in earthen pots.
 (B) LPG cylinders used for cooking.
 (C) Making salt from seawater.
 (D) Testing the purity of milk.
- 25.** Identify the specific tissue that forms the "hump" of a camel and serves as an insulator and energy reservoir.
 (A) Areolar tissue (B) Tendon
 (C) Adipose tissue (D) Cartilage
- 26.** A satellite orbits the Earth in a perfectly circular path. The work done by the Earth's gravitational force on the satellite is _____.
 (A) Positive, because it keeps the satellite in orbit.
 (B) Negative, because gravity pulls it inwards.
 (C) Zero, because the force is perpendicular to the displacement at every point.
 (D) Infinite, because the motion is continuous.
- 27.** The specific charge of a cathode ray is found to be _____.
 (A) Dependent on the nature of the gas in the tube.
 (B) Constant and independent of the gas or electrode used.
 (C) Maximum when the tube is filled with Nitrogen.
 (D) Zero because electrons are neutral.
- 28.** A farmer wants to develop a new variety of cattle that combines "high milk yield" with "high resistance to tropical diseases." Which of the following breeding strategies should he adopt?
 (A) Cross-breeding a local breed like *Red Sindhi* with another local breed like *Sahiwal*.
 (B) In-breeding within a foreign breed like *Jersey* for several generations.
 (C) Cross-breeding an exotic breed like *Brown Swiss* (high yield) with a local breed like *Sahiwal* (high resistance).
 (D) Random mating between any available cattle in the farm.
- 29.** A boy of mass 50 kg runs up a staircase of 45 steps in 9 seconds. If the height of each step is 15 cm, find his power. (Take $g = 10 \text{ m/s}^2$)
 (A) 375 W (B) 500 W
 (C) 337.5 W (D) 450 W
- 30.** You are given a mixture of sand and water. After leaving it undisturbed, the sand settles at the bottom. The process of pouring out the clear liquid without disturbing the sand is called _____.
 (A) Sedimentation (B) Decantation
 (C) Filtration (D) Evaporation
- 31.** While analyzing the muscular system, a student observes a tissue that is cylindrical, branched, and uninucleated. These cells also show rhythmic

contraction and relaxation throughout life. Where is this tissue located?

- (A) Attached to the humerus bone for movement.
- (B) In the wall of the heart.
- (C) In the lining of the small intestine.
- (D) In the iris of the eye.

32. A cube of side 10 cm is completely immersed in water. If density of water is 1000 kg/m^3 , buoyant force on the cube is: ($g = 10 \text{ m/s}^2$)

- (A) 10 N (B) 100 N
- (C) 1 N (D) 0.1 N

33. Why is the atomic weight of many elements, such as neon (20.18 grams per mole), expressed as a fractional value rather than a whole number?

- (A) Because of the mass of electrons.
- (B) Because atoms can be divided into smaller fragments.
- (C) Due to the presence of isotopes with different mass numbers.
- (D) Because the number of protons can vary in an element.

34. Rohan observes that while travelling in a lift going downward with increasing speed, he feels lighter on the weighing machine. If his actual weight is 480 N and the lift moves downward with acceleration 2 m/s^2 , what reading will the machine show? ($g = 10 \text{ m/s}^2$)

- (A) 384 N (B) 480 N
- (C) 576 N (D) 400 N

35. According to Bohr's model, what happens to the energy of an electron

when it moves from the L-shell to the K-shell?

- (A) Energy is absorbed.
- (B) Energy is released.
- (C) Energy remains constant.
- (D) The mass of the electron increases.

36. A truck carrying heavy cement bags has wide tyres, while a bicycle has thin tyres. If both push down on the road with the same force of 1200 N, and the truck tyre touches the road over 0.3 m^2 while the bicycle tyre touches only 0.02 m^2 , what is the ratio of the pressure exerted by the truck to that of the bicycle?

- (A) 1 : 15 (B) 15 : 1
- (C) 2 : 3 (D) 3 : 2

37. A solution of common salt is heated in a sand bath until only a white powder remains. This method of separation is suitable for ____.

- (A) Separating two liquids with different boiling points.
- (B) Removing an insoluble solid from a liquid.
- (C) Recovering a non-volatile soluble solid from its solvent.
- (D) Separating a mixture of two sublimable solids.

38. A fisherman uses a lactometer to check purity of milk. If lactometer sinks deeper than normal, it indicates:

- (A) Milk is pure
- (B) Density is lower due to added water
- (C) Milk has more fat
- (D) Temperature is very high

39. What is the electronic configuration of a phosphorus atom, which has an atomic number of 15?

- (A) $1s^2 2s^2 2p^6 3s^2 3p^2$
- (B) $1s^2 2s^2 2p^6 3s^2 3p^3$
- (C) $1s^2 2s^2 2p^6 3s^1 3p^4$
- (D) $1s^2 2s^2 2p^6 3s^2 3p^6$

40. A skydiver experiences increasing speed at first but later constant speed called terminal velocity because:

- (A) Gravity disappears
- (B) Air resistance balances weight
- (C) Mass reduces
- (D) g increases

41. If a metal 'X' is found to have 3 valence electrons, what is the most likely chemical formula of its oxide?

- (A) XO
- (B) X_2O
- (C) X_2O_3
- (D) X_3O_2

42. Imagine a hypothetical planet 'X' which has a mass double that of the Earth and a radius three times that of the Earth. If an object weighs 900 N on Earth, what would be its weight on planet 'X'?

- (A) 200 N
- (B) 100 N
- (C) 300 N
- (D) 450 N

43. Match the following physical and chemical changes with their corresponding examples:

Change Type	Example
1. Physical Change	(i) Rusting of iron
2. Chemical Change	(ii) Melting of wax
3. Both Physical & Chemical	(iii) Burning of a candle

Which of the following represents the correct matching?

- (A) $1 \rightarrow (ii)$, $2 \rightarrow (i)$, $3 \rightarrow (iii)$
- (B) $1 \rightarrow (i)$, $2 \rightarrow (ii)$, $3 \rightarrow (iii)$
- (C) $1 \rightarrow (iii)$, $2 \rightarrow (i)$, $3 \rightarrow (ii)$
- (D) $1 \rightarrow (ii)$, $2 \rightarrow (iii)$, $3 \rightarrow (i)$

44. A heavy wooden box is pushed on a rough floor with a force of 40 N, but it does not move. When the applied force is increased to 60 N, the box starts moving. Which statement best explains this observation?

- (A) Friction increases with applied force
- (B) Balanced forces act below 60 N and become unbalanced beyond it
- (C) The box has greater inertia at lower forces
- (D) Motion starts only when force exceeds mass

45. **Assertion (A):** The density of a liquid is generally greater than that of its solid state.

Reason (R): In liquids, molecules are loosely packed with more intermolecular space compared to solids.

- (A) Both A and R are true, and R is the correct explanation of A.
- (B) Both A and R are true, but R is not the correct explanation of A.
- (C) A is true, but R is false.
- (D) A is false, but R is true.

46. Two spheres, P and Q, are made of the same material. The radius of sphere Q is twice the radius of sphere P. If both spheres are subjected to the same amount of unbalanced force, what will

be the ratio of the acceleration of sphere P to that of sphere Q?

- (A) 1:2 (B) 4: 1
(C) 8:1 (D) 1:8

47. A sample of milk is rotated in a machine at a very high speed, causing the lighter cream to float on top of the heavier contents. This method of separation is called _____.

- (A) Filtration (B) Sublimation
(C) Centrifugation (D) Sieving

48. A force of 10 N acts on a body of mass 2 kg for 5 seconds. Another force of 20 N acts on a body of mass 4 kg for the same time. Which statement is correct?

- (A) Both bodies gain equal acceleration
(B) Both bodies gain equal momentum
(C) First body gains more velocity
(D) Second body gains less momentum

49. A wooden block of mass 5 kg is being pushed against a vertical wall by a horizontal force of 100 N. If the coefficient of static friction between the block and the wall is 0.6, what is the frictional force acting on the block? (Take $g = 10 \text{ m/s}^2$)

- (A) 60 N (B) 100 N
(C) 50 N (D) 0 N

50. Two billiard balls, each of mass 0.05 kg, moving in opposite directions with speed 6 m/s collide and rebound with the same speed. What is the impulse imparted to each ball?

- (A) 0.3 kg m/s (B) 0 kg m/s
(C) 0.6 kg m/s (D) 0.06 kg m/s

KEY

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|-------|-------|-------|-------|-------|-------|-------|
| 1) C | 2) B | 3) B | 4) B | 5) C | 6) C | 7) B |
| 8) D | 9) C | 10) A | 11) A | 12) C | 13) A | 14) A |
| 15) B | 16) C | 17) A | 18) B | 19) C | 20) D | 21) C |
| 22) C | 23) C | 24) B | 25) C | 26) C | 27) B | 28) B |
| 29) A | 30) B | 31) B | 32) A | 33) C | 34) A | 35) B |
| 36) A | 37) C | 38) B | 39) B | 40) B | 41) C | 42) A |
| 43) A | 44) B | 45) D | 46) C | 47) C | 48) A | 49) C |
| 50) C | | | | | | |